AlphaDecay: Autonomous Financial Research Agent

System Architecture, Two-Agent Fact-Checking & Financial Calculations

1. Executive Overview: Autonomous Discovery vs. Summarisation

Most financial AI tools just summarize news articles and often make math mistakes when calculating financial ratios. AlphaDecay is an autonomous financial research agent built for the ATMOS 2026 Hackathon ('Finding the Next Signal'). Given any NIFTY 50 FMCG company from Track C, AlphaDecay downloads public financial filings, calculates audited accounting ratios in Python, forms an original hypothesis, challenges it with a skeptical disproving agent, and produces an evidence-backed research deliverable meeting all 8 hackathon brief requirements.

2. Autonomous Investigation Process Workflow (6 Phases)

Phase	Engine Component	Operational Execution & Verification Details
Phase 1: Ingestion	Data Fetchers (Screener & BSE)	Downloads multi-year quarterly P&L;, annual balance sheets, and cash flow statements from audited SEBI LODR filings; extracts dialogue blocks from official BSE concall PDFs via PyPDF.
Phase 2: Math Engine	Pure Python Engine (analytics.py)	100% pure Python execution. Enforces strict period matching: annual CFO benchmarked against audited annual EBITDA ex-Other Income; quarterly sales YoY matched to identical quarter t-4. No AI math mistakes.
Phase 3: Triage	Divergence Scoring (analytics.py)	Ranks observed divergences into candidate anomalies by severity (/10): CFO-to-EBITDA decoupling, supplier payable stretches, DSO debtor velocity shifts, inventory DIO expansions, and margin changes.
Phase 4: Fact-Checking	Two-Agent Loop (pipeline.py)	Investigative Agent formulates a causal hypothesis. Disproving Agent acts as skeptical defense, testing whether ordinary FMCG commercial dynamics explain the shift. If refuted, Attempt 2 refines the thesis.
Phase 5: Adjudication	Verification Gate	Emits clear binary verdicts: HYPOTHESIS PASSED (withstood scrutiny) or HYPOTHESIS FAILED (refuted by business context). Synthesizes the official 8-point research brief.
Phase 6: Delivery	ReportLab PDF & Web Dashboard	Renders a clean 2-page dark research report, dual-axis trajectory chart, and interactive web dashboard featuring live execution polling, audited formulas, and direct BSE clickable links.

3. Multi-Agent Architecture: The Nodes & Two-Agent Debate

AlphaDecay utilizes a specialized multi-agent graph with distinct roles. To eliminate confirmation bias, no single agent is allowed to declare a finding without a skeptical defense cross-examination:

Agent / Node	Role	How It Works & What It Prevents
Investigative Agent (Prosecution)	Formulates causal hypotheses connecting mathematical divergences to operational anomalies.	Analyzes top-ranked anomaly candidates against concall transcripts. Pinpoints non-obvious operational explanations (e.g., vendor payable optimization, marketing accrual cash drain).
Disproving Agent (Defense)	Skeptical stress-tester. Seeks to refute the hypothesis using normal commercial context.	Challenges hypotheses using structural FMCG realities (e.g. negative CCC of -89 days proving supplier leverage, raw material price buffering). Breaks down ungrounded claims.
Cyclic Refinement (Attempt 1 -> 2)	Self-correcting iterative retry loop across anomalous candidates.	If Attempt 1 is refuted by the defense, the agent synthesizes defense critique to formulate an airtight Attempt 2, or abandons the thesis and marks it as refuted.
Adjudication Node (Independent Judge)	Enforces clear binary verdicts and audits evidence strength.	Evaluates whether the defense broke down the hypothesis. Assigns binary HYPOTHESIS PASSED or HYPOTHESIS FAILED, verifies evidence trail, and frames falsification criteria.

4. How Financial Metrics & Formulas Are Calculated

To guarantee mathematical accuracy, arithmetic is kept entirely separate from LLM text generation. All indicators are calculated directly in pure Python using audited balance sheets and disclosures:

Financial Metric	Exact Formula	Period Alignment	What This Tells You
CFO-to-EBITDA Ratio	`(Audited_CFO / Audited_Operating_Profit) * 100`	Audited Annual to Annual	Earnings quality test. A drop below 80% (e.g. HUL: 80.9% to 73.1%) reveals operating profit decoupling from cash collected in the bank.
EBITDA (ex-Other Inc)	`Net_Sales - Total_Operating_Expenses`	Quarterly & Annual	Isolates pure manufacturing/distribution operating profit, stripping non-operating treasury income gains.
Cash Conversion Cycle	`DSO (Debtor) + DIO (Inv) - DPO (Payable)`	Audited Annual	Measures working capital financing. A deeply negative cycle (-89d for HUL) proves vendors are funding operational inventory.
Debtor / DSO Velocity	`(Trade_Receivables / Audited_Sales) * 365`	Annual Audited	Distributor credit terms. Measures how fast wholesale trade partners settle receivables (HUL: improved 23d -> 19d).

AlphaDecay // Technology Stack, Development Timeline & Deliverable Compliance

REPOSITORY: github.com/aryanmsrh/alphadecay

5. Verified Technology Stack & Implementation Details (Audited Against Codebase)

Layer	Actual Technology Used	Codebase Verification & Architectural Implementation Details
Core & Runtime	Python 3.10+, Flask 3.1, Atomic JSON Store	Pure Python rule-based calculation core; lightweight Flask microservice (run_web.py); transactional JSON run store (webapp/run_store.py) with atomic .tmp swap.
LLM & Reasoning	Google Gemini 3.5 Flash Lite + Ollama (DeepSeek-R1)	Direct HTTP client (httpx) with high thinking level; cloud Gemini 3.5 Flash Lite primary; local Ollama deepseek-r1:7b offline reasoning fallback.
Ingestion & Harvest	Requests, BeautifulSoup4, PyPDF, Tavily API	Screener.in scraping via requests/BS4; official BSE concall PDFs downloaded via httpx and parsed via PyPDF; targeted financial search via Tavily.
Exporters & Visuals	ReportLab 5.0, Matplotlib 3.11, Chart.js	ReportLab custom flowables for strictly 2-page dark research PDF; Matplotlib dual-axis sales vs margin chart; Chart.js in browser workstation.
Frontend UI	Dark Web Dashboard (#070709)	Space Grotesk + Plus Jakarta Sans + JetBrains Mono typography; responsive calculation cards; micro-elevation shadows.

6. Development Timeline: Day-by-Day Engineering Milestones (16th – 19th Oct 2026)

Date & Phase	Milestone Target	Key Technical Hurdles Solved & Architecture Built
16 October 2026 Day 1: Foundations	Problem Framing & Data Ingestion Scaffolding	Scoped Track C (Consumer, FMCG & Retail). Built robust Screener.in scraper parsing audited quarterly P&L;, annual balance sheets, and cash flows. Engineered BSE India concall PDF downloader and dialogue extractor.
17 October 2026 Day 2: Python Calculations	Pure Python Analytics & Period Alignment	Eliminated LLM math mistakes by writing pure Python formulas for YoY revenue growth, EBITDA ex-other income, CFO conversion, DIO, DSO, and CCC. Resolved quarterly-vs-annual mismatch by locking audited annual period baselines.
18 October 2026 Day 3: Two-Agent Pipeline	Two-Agent Loop & ReportLab Exporter	Architected the 2-attempt adversarial challenge loop (Investigative Agent vs Disproving Agent). Built binary adjudication verdict logic (HYPOTHESIS PASSED / FAILED). Authored custom ReportLab canvas strictly constrained to 2 pages.
19 October 2026 Day 4: Polish & Packaging	Web Dashboard, Bug Fixes & Live Testing	Redesigned UI to pitch-black workstation (#070709) with Space Grotesk / JetBrains Mono typography. Fixed double-sign (+- / ++) artifacts and hardcoded significance strings. Tested clean run on HUL with verified 2-page PDF and merged to main.

7. Official 8-Point Hackathon Deliverable Brief Compliance

Brief Requirement	AlphaDecay Implementation	Brief Requirement	AlphaDecay Implementation
1. The Discovery	One clear sentence describing the non-obvious finding.	5. Numbers	Accurate financial calculations & formulas substantiating finding.
2. Why It Matters	3 to 5 bullets detailing operational & capital significance.	6. Counterargument	Strongest alternative explanation evaluated by defense agent.
3. Evidence	Multiple independent sources with direct, clickable links.	7. Falsification	Explicit future empirical condition that would disprove finding.
4. Timeline	Chronological emergence across sequential reporting periods.	8. Evidence Strength	Strong, Moderate, or Weak rating with audit rationale.

8. Key Architectural Advantages

1. Accurate Math: Python calculates all numbers directly, eliminating math errors during evaluations.
2. Generalizes: Works out of the box for any listed Track C company (HUL, ITC, Nestle, Britannia, Tata Consumer).
3. Offline Model Fallback: Easily switches to local Ollama (deepseek-r1:7b) via --offline if cloud keys are unavailable.
4. Auditable Evidence: Every finding links back to official SEBI filings and BSE transcripts so claims can be verified immediately.